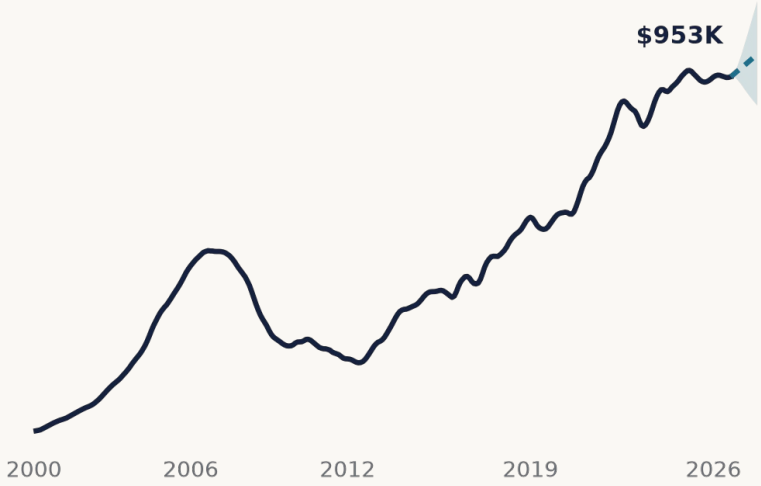
MGMTMSA 437 · FINAL PROJECT · OCTOBER 2026

Forecasting LA Home Values

Mortgage rates more than doubled after 2021. Did LA home values get harder to predict?

Yang Liu · Jinwoo Roh
Yiu-Hon Lee · Thanapol Doungsaeng

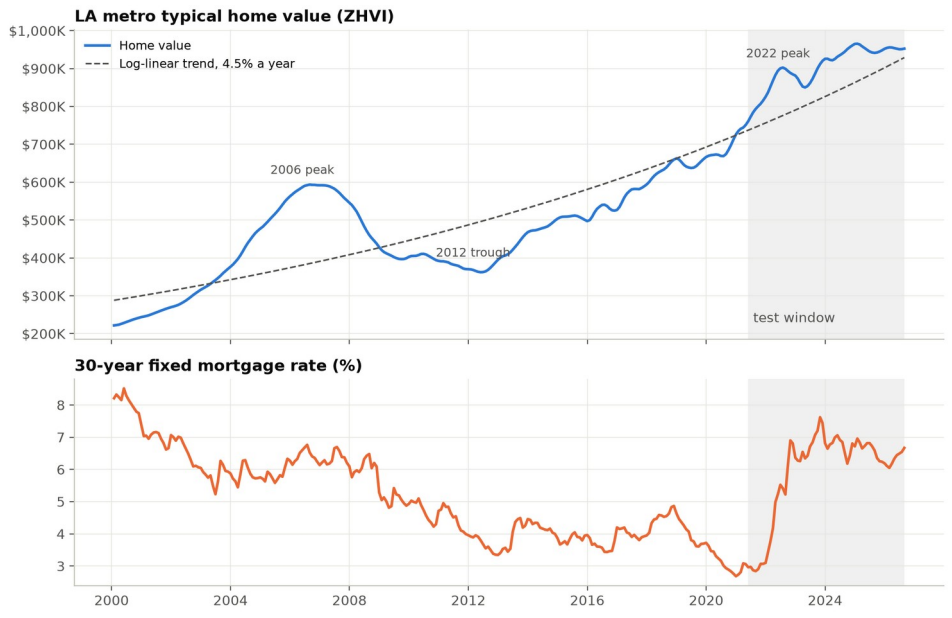
Zillow Home Value Index, LA metro, Jan 2000 to Aug 2026, plus 499 neighborhoods and the 30-year mortgage rate



Solid: actual · Dashed: our forecast to Aug 2027

THE QUESTION

LA values grew 5.6% a year for 26 years, then stalled when rates hit 7%



$952,601

typical LA-metro home, August 2026

-1.3%

vs the January 2025 peak of $965,634

2.7% → 7.6%

30-year mortgage rate, 2021 low to 2023 high

DATA AND DESIGN

We tested 14 models two ways, so no model could win by luck

DATA

Zillow ZHVI, LA metro
320 months, Jan 2000 to Aug 2026

499 LA neighborhoods
Zillow neighborhood index

Mortgage rate and payroll jobs
FRED and BLS, LA and Orange County

14 MODELS + 3 LLMS

Baseline and classical
Naive drift · ARIMA · Holt · Prophet ×2

ML, deep learning, AI
XGBoost · LightGBM · LSTM · GRU · Chronos

Outside information
ARIMAX ×2 · VAR · factor VAR

Plus 3 chat LLMs
Claude · GPT · Gemini

2 TESTS

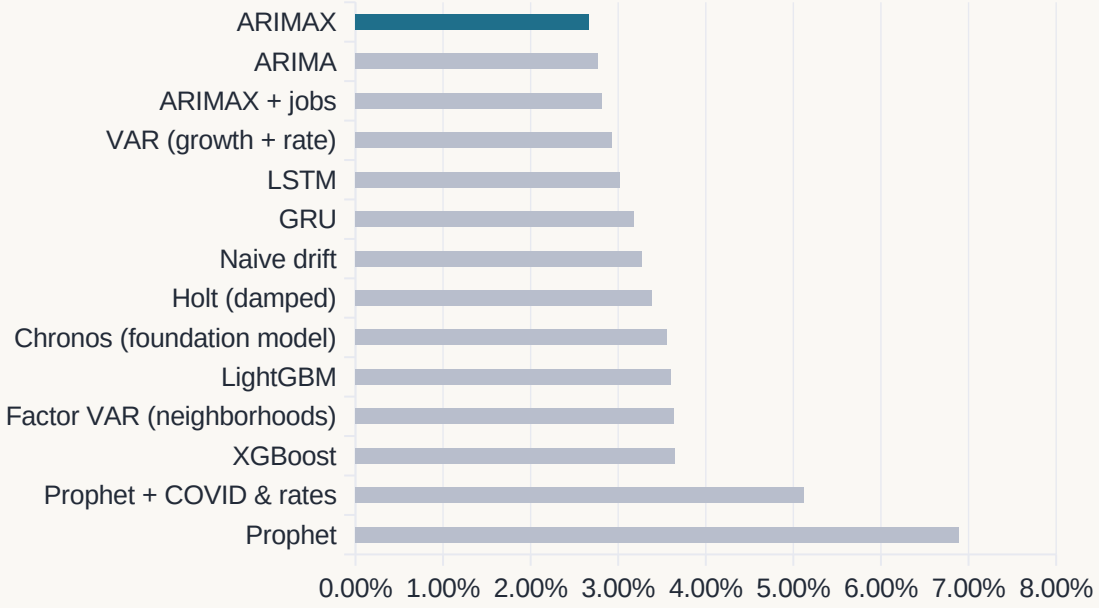
Blind 80/20
Train to Apr 2021, forecast 64 months in one shot

Rolling origin
Refit every 3 months, forecast 12 months ahead, 18 times

WHY NOT 1-MONTH ERRORS Zillow smooths this index, so every decent model is about 0.1% off one month ahead. We judge models at 6 to 12 months instead.

RESULT 1 · ROLLING TEST

In the fair test, ARIMA-family models beat machine learning



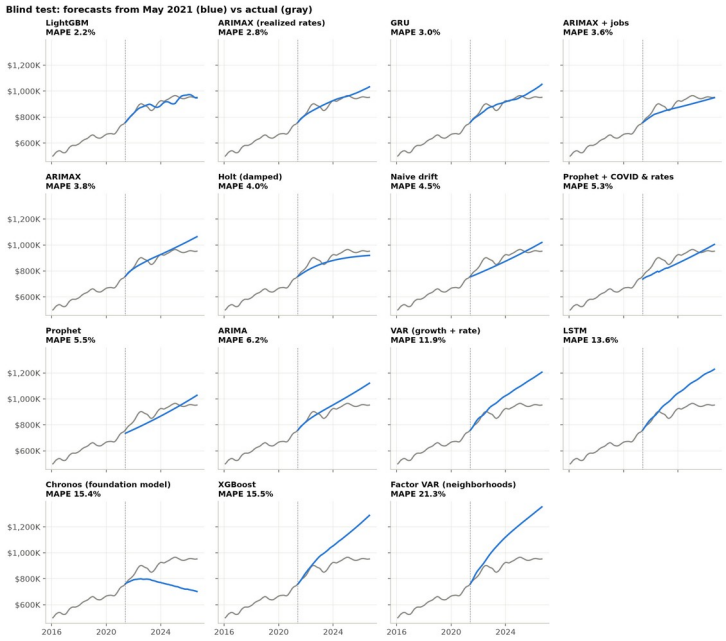
LOWEST ERROR

2.67%

ARIMAX, average error 1 to 12 months ahead. ARIMA 2.77%. LSTM 3.02%, XGBoost 3.65%, Prophet 6.89%. Also 1st on RMSE ($35.8K).

RESULT 2 · SINGLE BLIND TEST (ONE WINDOW)

One test window crowned LightGBM; 18 windows ranked it tenth



SAME MODEL, TWO TESTS

1st → 10th

LightGBM won the single 64-month test (2.2% error) but ranked 10th across 18 rolling tests: it got lucky in the one window that counted.

RESULT 3 · AI FORECASTS

Chat LLMs won by memory, not forecasting



SAME MODEL, DATES HIDDEN

1.3 → 2.8%

Claude's error once it could not tell the place, dates or mortgage rate (ARIMAX 2.67%). All test dates are inside the LLMs' training data.

RESULT 3 · AI FORECASTS, HOW WE TESTED

The LLMs' edge disappears once we mask the data

NAMED

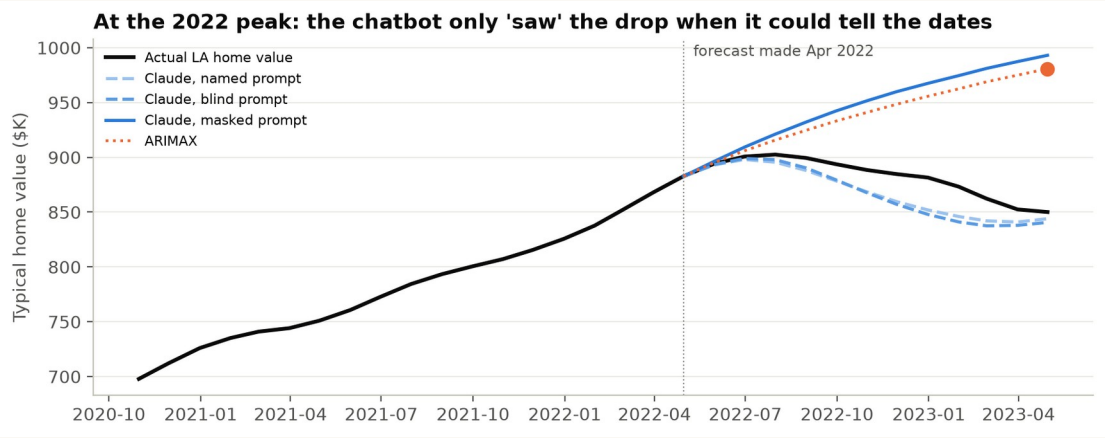
LA, real dates, dollars, mortgage rate
Leaks: it knows the place and the years

BLIND

No place or dates, but the real rate
Leaks: the 2022 rate jump dates the series

MASKED

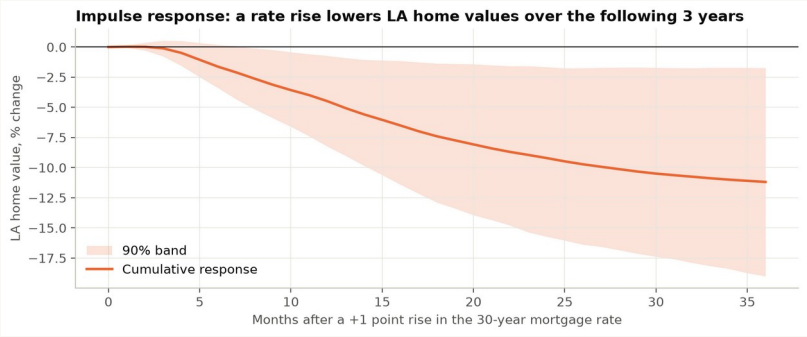
Last 10 years only; no place, dates or rate
The fair test

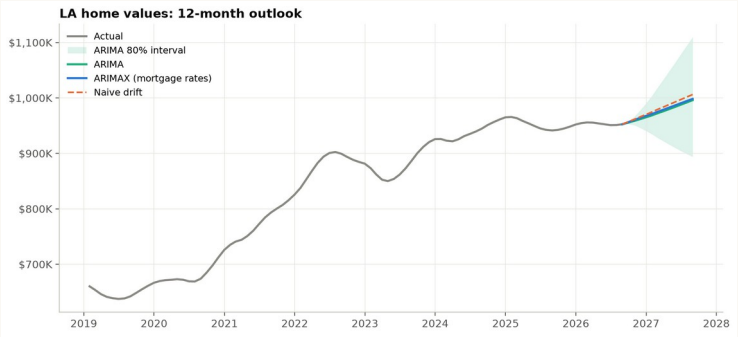


NAME THIS SERIES Given the blind data, Claude named Los Angeles 16 of 18 times and got the dates exactly right. Masked: 0 of 18.

RESULT 4 · MORTGAGE RATES AND OUTLOOK

Rates act slowly; we forecast +5% by Aug 2027





−9.2%

two years after a 1-point rate rise (−4.5% after one)

p = 0.052

rates lead prices (Granger, borderline). In ARIMAX: rate p = 0.16, jobs p = 0.29

$998K (+4.8%)

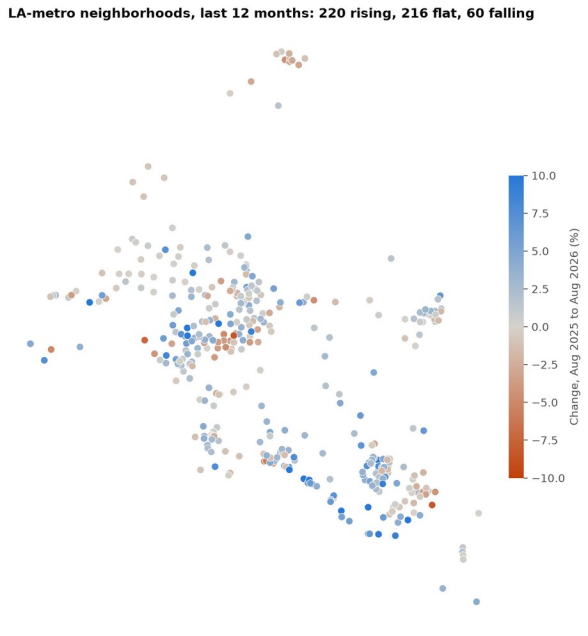
ARIMAX forecast, August 2027

-6% to +17%

80% range for the next 12 months

NEIGHBORHOODS · LIVE MAP

The metro average hides a split market, neighborhood by neighborhood



220 / 216 / 60

neighborhoods rising / flat / falling more than 2%, last 12 months

152 of 496

still below their 2022 peak, mostly in LA County

−1.2 pts

deeper 2022 drop per doubling of price in LA County (reverses in Orange County)

Live map: neiltd.github.io/la-housing-forecast

TAKEAWAYS

What we learned, and what we would do next

01

Simple beats complex, if you test fairly

ARIMA models beat ML and Chronos; one blind window would have crowned LightGBM (10th overall).

02

An LLM can't be backtested on years it has read about

Chatbots beat every model until we hid the place, dates and rate; then they only tied ARIMAX.

03

Mortgage rates matter, slowly; local jobs don't add

A 1-point rise lowers values about 9% over two years. Turning points stay hard to call.

04

The metro average hides a split market

Separate models per neighborhood beat one shared factor (2.8% vs 4.1% error).

NEXT STEPS Test more cities; score the AI live on months after its training cutoff; try made-up look-alike series the AI cannot have seen.